

THE PROBLEM

Enterprise teams waste **30-45 minutes daily** hunting for information locked in department silos. Finance can't access HR headcount for budget planning. New employees don't know where anything lives. Critical documents exist — but the wrong people can't find them, and the right people can't control who sees what.

Siloed data

No access control

Manual search

Compliance risk

THE SOLUTION

A **secure, role-aware AI chatbot** that gives every employee instant answers from company documents — but only documents their role is authorized to see. Every answer is grounded in real company data with source citations. No hallucinations. No data leaks.

Role-based access

Source citations

On-premise LLM

Audit logs

HOW IT WORKS

1	2	3	4	5	6
Login	Ask	RBAC Check	Vector Search	AI Answer	Cite Sources
Employee signs in — JWT token issued with their role	Natural language question typed into the chat UI	System verifies role before any data is retrieved	Semantic search across authorized documents only	LLM generates answer grounded in retrieved context	Every claim linked to the source document + page

TECH STACK

Layer	Technology	Purpose
LLM	Llama 3.2 (local)	Chat generation — no data leaves company
Embeddings	nomic-embed-text	Semantic document search
Vector DB	Qdrant	Role-filtered document retrieval
Backend	FastAPI (Python)	REST API + streaming responses
Frontend	Streamlit	Role-aware chat interface
Auth	JWT + bcrypt	Secure login + session management
Database	PostgreSQL / SQLite	User & audit data
Deploy	Docker + AWS/GCP	Containerised, cloud-ready

WHO BENEFITS

Role	Key Benefit
Finance	Instant P&L, budgets, forecasts — no
HR	Policy Q&A 24/7 — frees HR team for
Engineering	Architecture docs & runbooks always f
Marketing	Campaign data & brand guidelines on
C-Level	Cross-dept intelligence in one convers
All Staff	Policies, expenses, holidays answered

KEY DIFFERENTIATORS vs GENERIC AI TOOLS

Feature	FinSolve AI	Generic ChatGPT / Copilot
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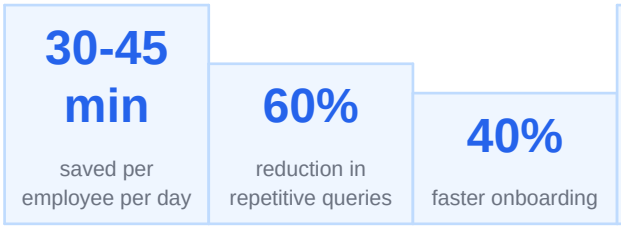
Data source	Your company docs only	General internet knowledge
Access control	Role-based — built in at core	None / bolt-on
Hallucinations	Prevented — context-only answers	Common risk
Data privacy	100% on-premise capable	Data sent to external API
Source citations	Every answer cited	Rarely provided
Compliance	Full audit log per query	No audit trail
Customisation	Department-specific personas	One-size-fits-all

TIMELINE — 6 TO 8 WEEKS

Phase	Weeks	Deliverable
Auth + RBAC	1-2	Login, roles, user management
RAG Pipeline	3-4	Document ingestion + vector search
Chat UI	5-6	Streaming UI + source citations
Test + Deploy	7-8	Security audit + production deploy

1-2 engineers. Expandable to multi-tenant with minimal refactoring.

EXPECTED IMPACT



Scalable from day one — stateless API, distributed vector DB, swappable LLM provider. Handles 500+ concurrent users at under 4s response time.

SCALABILITY

Horizontal Scale

Stateless FastAPI scales behind any load balancer

Vector DB Sharding

Qdrant supports millions of documents distributed

Swappable LLM

Switch Llama to GPT-4o with one config change

Multi-Tenant

Same architecture supports multiple business units